Georgia State University
CSC 4210/6210 Computer Architecture
Spring Semester 2026

Processor Design Semester Project - Task4:

Release Date: 04-10-2026
Due Date: 04-20-2026

Single-Cycle Processor Design (AND / OR)

You are a CPU architect at Intel, California. You are asked to design a single cycle 32-bit processor. A single-cycle processor datapath that executes an instruction set at Intel. The processor must support basic logic operations to evaluate a Boolean expression.

The target computation is: Y = A·B + C'·D

Section 1 — Instruction Set & Program:

1. Your processor must support the following instructions:
   a. AND
   b. OR
2. The NOT operation must be supported using:
   a. ALU input inversion (controlled via control signal)
   b. NOT must not be implemented as a separate instruction
3. The processor must execute the following program:

```
Assuming: t0=A, t1=B, t2=C, t3=D
and t4, t0, t1    ; t4 = A & B
and t6, t5, t3    ; t6 = (~C) & D
or  t0, t4, t6    ; t0 = t4 | t6
```

4. The inversion of C must be handled using:
   a. ALU control signal (e.g., invert input flag)
   b. Encoded in function field (not opcode)

Section 2 — Datapath Design:

1. Implement a single-cycle datapath consisting of:
    a. Register file
    b. ALU
    c. Multiplexers (MUX)
    d. Instruction input
    e. Control signals
2. Each datapath component must be implemented as a separate programming file
3. The datapath must support:
    a. Register read (two inputs)
    b. Register write (one output)
    c. ALU operation selection (AND / OR)
    d. ALU input inversion (for NOT functionality)

Section 3 — Control Unit:

1. Implement a control unit that:
    a. Decodes instruction fields
    b. Generates control signals
2. Control signals must include:
    a. ALU operation (AND / OR)
    b. Inversion flag (for NOT behavior)
    c. Register write enable
3. The control logic must differentiate between:
    a. Standard AND
    b. AND with inversion (AND-not behavior)

Section 4 — Execution Model:

1. The processor must follow a single-cycle execution model:
    a. Fetch → Decode → Execute → Write-back in one cycle
2. All operations must complete within one cycle (logical simulation)

Section 5 — Validation:

1. Provide input values for:
    a. A, B, C, D
2. Execute the program and verify:
    a. Final output matches:

$$Y = A \cdot B + C' \cdot D$$

3. Show intermediate register values:
    a. t4 = A & B
    b. t6 = (~C) & D
    c. t0 = final result

Program Output:

1. Instruction execution trace
2. Control signals per instruction
3. Register values after each instruction
4. Final output (Y)

Deliverables:

1. Source code (modular: each datapath stage separate file)
2. README with instructions
3. Submit code to github
4. 4-5 minute demo video explaining code and displaying execution
    a. Either upload it to iCollege or storage of your choice with necessary viewing permissions
5. Can video recordings be uploaded to youtube? Yes or No